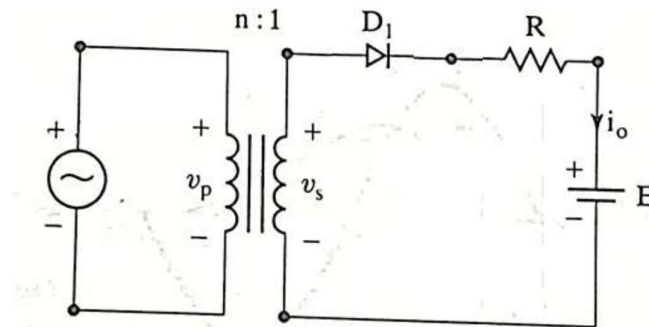


1-phase and 3-phase half and full wave full bridge diode rectifier with R and RL loads

Q1. A DC motor load is connected through a three-phase half wave rectifier with a Y connected 120V (per phase) 60 Hz 3-phase source as the input. The internal resistance of the motor is 6Ω . Calculate (a) the average output voltage (b) average current through the diode (c) PIV of each diode (d) FF of output voltage (e) RF of output voltage (f) Crest factor of output voltage, if the current is ripple free.

Q2. A three-phase bridge rectifier has an RL load with very large inductance providing a ripple free current, such that the average load current is 60A, when fed from a 120V, 50Hz supply. Determine the average output voltage, PIV of the diodes, input line current RMS value and R load.

Q3. The battery voltage for the circuit shown in figure is given as 12V and its capacity is 100Whr. The average charging current should be 5A. The voltage provided to the primary side of the transformer is 120V (RMS), 60Hz and the transformer has a turns ratio of 2:1. Calculate (a) the conduction angle of the diode using suitable waveforms and clearly marking all the values, (b) the current limiting resistance, R, (c) the power rating of the limiting resistance, (d) the charging time in hours, (e) the rectifier efficiency and the (f) the PIV of the diode.



Q4. A full wave bridge rectifier has an AC source with V_m (peak) = 100V at 60 Hz and a series RL load (a) With $R = 10\ \Omega$, and $L = 10\text{mH}$ in the continuous current mode, determine the (i) average current in the load (ii) power dissipated as heat in the resistor (b) With $R = 10\ \Omega$, and very large inductance in the continuous current mode, determine the (i) average diode current (ii) RMS diode current